

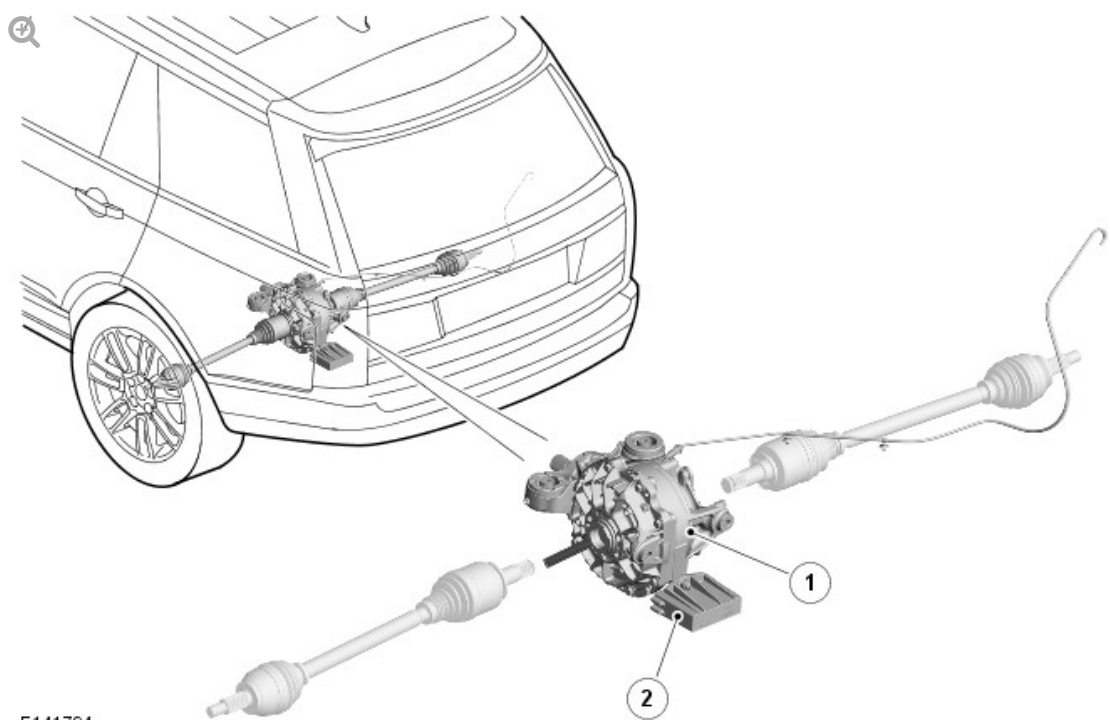
2016.0 RANGE ROVER (LG), 205-02

REAR DRIVE AXLE/DIFFERENTIAL

DESCRIPTION AND OPERATION

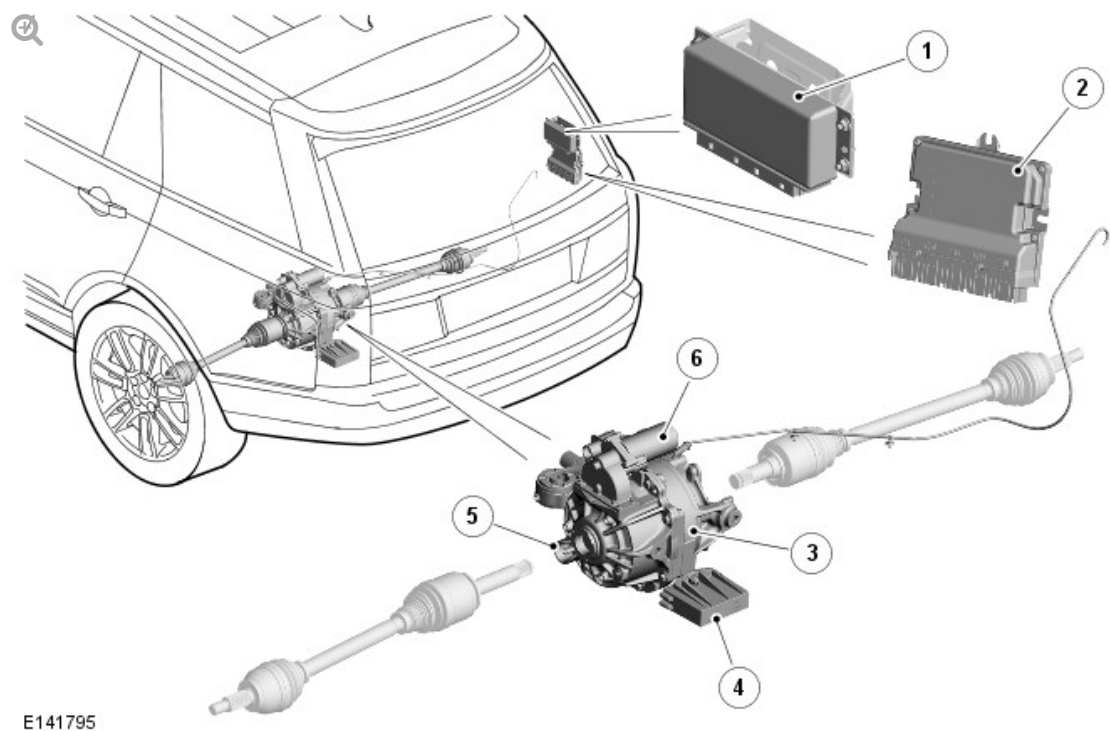
COMPONENT LOCATION

Open Differential



ITEM	DESCRIPTION
1	Open differential
2	Damper

Electronic Differential



ITEM	DESCRIPTION
1	Rear differential control module
2	Integrated suspension control module
3	Electronic differential
4	Damper
5	Oil temperature sensor
6	Differential locking motor

OVERVIEW

The differential has two functions:

- to convert the 'angle of drive' through 90° and distribute drive, via the rear drive halfshafts, to the rear wheels.
- to compensate for differences in the rotational speeds of the vehicle's rear wheels during cornering.

Depending on vehicle specification there are two types of differential installed:

- open differential
- electronic differential.

Both types of differential are attached to the rear subframe at four mounting points. Each mounting point incorporates a rubber bush to reduce NVH (noise, vibration and harshness). The bushes in the forward mounting points are installed in the differential. The bushes in the rear mounting points are installed in the rear subframe. A damper weight attached to the rear of the differential is used to control torsional vibrations.

DESCRIPTION

OPEN DIFFERENTIAL

The open differential is a conventional design using a hypoid gear layout. For additional information, refer to: Specifications (205-02 Rear Drive Axle/Differential, Specifications).

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The cast iron casing comprises two parts; a cover and a carrier. The carrier provides locations for all the internal components. The cover is sealed to the carrier with Loctite RTV sealant and secured with bolts. The cover and carrier have cast fins, which assist rigidity and cooling. A breather cap is fitted to the top of the carrier.

The open differential contains a quantity of oil for splash lubrication of the internal components. A magnetic drain plug is installed in the bottom of the carrier and a filler/level plug is installed in the cover.

The pinion shaft has an externally splined end which offers a direct connection to the driveshaft via the DTF (Direct Torque Flow) connection. In comparison to the conventional flange connection the DTF connection provides:

- Reduced weight due to fewer components.
- Smaller outer diameter.
- Improved driveshaft imbalance.
- Reduced assembly time.

The pinion shaft has a hypoid gear at its inner end, which mates with the crown wheel drive gear.

The crown wheel drive gear is located on the differential carrier and secured with bolts. The differential carrier is mounted on taper roller bearings located in machined bores in the carrier and the cover. Shims are installed behind the bearing cups to apply the correct bearing preload and hypoid backlash.

The differential carrier has a through hole, which provides location for a cross shaft. The planet gears are installed on the cross shaft, with thrust washers between the planet gears and the differential carrier. A roll pin locks the cross shaft to the differential carrier.

The sun gears are located in pockets in the differential carrier and mesh with the planet gears. Belleville washers are fitted between the sun gears and the differential carrier and set the correct mesh contact between the planet gears and the sun gears. Each sun gear has a machined bore with internal splines and a machined groove. The splines transfer drive to the rear drive halfshafts. The groove provides positive location for the snap ring fitted to the inboard end of the rear drive halfshafts.

Oil seals are installed in the carrier and the cover to seal the rear drive halfshafts.

ELECTRONIC DIFFERENTIAL

The electronic differential has the same functionality as the open differential, but it also incorporates a locking and torque biasing function to give improved traction performance and vehicle dynamic stability. Operation of the electronic differential is controlled by the RDCM (Rear Differential

Control Module).

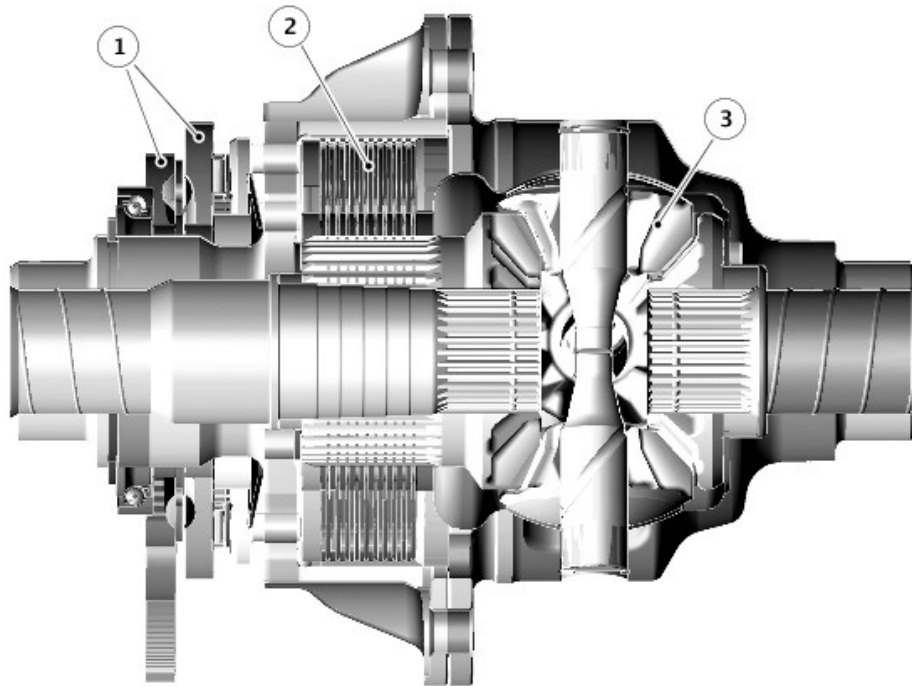
The basic construction of the electronic differential is similar to the open differential. However, the electronic differential also has the following:

- Two additional planet gears in the differential carrier, to cater for the higher torque through the differential during locking events.
- A multi-plate clutch and actuator assembly installed on the left-hand sun gear.
- A motor and reduction gearbox, attached to the cover.
- An oil temperature sensor installed in the cover.

The RDCM operates the motor of the electronic differential under the control of the Integrated Suspension Control Module.

The multi-plate clutch is contained in a clutch basket attached to the differential carrier with the crown wheel securing bolts. Alternate plates of the clutch pack are keyed to the clutch basket and the left-hand sun gear. A pressure disc is installed on the outer end of the clutch pack and keyed to the clutch basket. A thrust race on the end of the clutch basket incorporates lugs which extend through the clutch basket onto the pressure disc.

The actuator assembly is mounted on bearings on the outboard end of the clutch basket, against the thrust race. The actuator assembly consists of input and output actuators separated by five ball bearings. A locking pin in the cover engages with a slot in the output actuator to prevent it turning, but allow it to move axially. The input actuator engages with the reduction gearbox and is free to rotate relative to the cover. Ball bearings locate in curved grooves in the mating faces of the input and output actuators. The bottom surface of each groove incorporates a ramp. Rotation of the input actuator forces the ball bearings up the ramps in the grooves and induces an axial movement in the output actuator. The thrust race and pressure disc transfer the axial movement from the output actuator to the clutch pack.



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ITEM	DESCRIPTION
1	Actuator
2	Multi-plate clutch
3	Differential

The motor is a 12 V dc motor that adjusts the frictional loading of the multi-plate clutch, via the reduction gearbox and the actuator assembly, under the control of the RDCM. Adjusting the frictional loading of the multi-plate clutch adjusts the locking torque between the crown wheel drive gear and the sun wheel.

Four bolts attach the motor to the reduction gearbox, which is located in position on the cover with two dowels, and secured with four bolts. An O-ring seals the joint between the motor and the reduction gearbox.

The motor is driven by a 12 V dc feed direct from the RDCM. The motor also incorporates the following connections with the RDCM:

- A motor temperature sensor, to prevent excessive use from damaging the motor.

- Two Hall effect motor position sensors, to enable closed loop control of the motor.

The temperature sensor provides a differential oil temperature signal to the RDCM, to prevent excessive use from damaging the multi-plate clutch.

REAR DIFFERENTIAL CONTROL MODULE

The RDCM (Rear Differential Control Module) controls operation of the electronic differential. The RDCM is attached to a bracket located on the right-hand side of the luggage compartment.

The RDCM receives three battery feeds from the RJB (rear junction box) and an ignition feed from the CJB (central junction box). A connection with the high-speed CAN (controller area network) chassis bus allows the RDCM to communicate with other systems on the vehicle.

A certain amount of differential slip is required to allow the vehicle to turn corners and to remain stable under control of the ABS (anti-lock brake system). The Integrated Suspension Control Module (ISCM) monitors the driver's demands through primary vehicle controls and automatically sets the slip torque in the differential. The system is completely automatic and does not require any special driver input.

The differential strategy in the ISCM includes:

- A pre-loading function, increasing locking torque with increased driving torque.
- A slip controller to decrease locking torque for optimum comfort, for example when parking.

The ISCM memorizes the position of the motor when the ignition is switched off.

CAN chassis bus messages used by the ISCM include wheel speed, steering angle, automatic transmission speed, temperature information, car configuration, axle ratios and mode inputs.

The ISCM also sends messages via the CAN chassis bus to tell other control modules on the network the status of the electronic differential. The clutch torque and default mode status are some of the main signals sent out by the ISCM.

If the RDCM or ISCM are replaced, a Land Rover approved diagnostic system must be connected to the vehicle and the differential self-calibration procedure must be performed. This procedure must also be performed if the motor or electronic differential is replaced.

If a fault occurs with the electronic differential, the ISCM, the RDCM, or one of the required input signals, the ISCM records an error code and displays a warning in the message center.

The following messages can be displayed:

MESSAGE	DESCRIPTION	CHIME
E-DIFF NOT AVAILABLE	Differential temperature has reached the overheat threshold. System deactivated until temperature returns within limits.	Single
E-DIFF FAULT	Fault has occurred with electronic differential. System deactivated until fault rectified.	Single

OPERATION

OPEN DIFFERENTIAL

Rotational input from the drive shaft is passed via the input flange to the pinion shaft and pinion gear. The angles of the pinion gear to the crown wheel drive gear moves the rotational direction through 90°.

The transferred rotational motion is now passed to the crown wheel drive gear, which in turn rotates the differential casing. The cross-shaft, which is secured to the casing, also rotates at the same speed as the casing. The planet gears, which are mounted on the shaft, also rotate with the casing. In turn, the planet gears transfer their rotational motion to the left and right hand sun gears, rotating the drive halfshafts.

When the vehicle is moving in a forward direction, the torque applied through the differential to each sun gear is equal. In this condition both drive halfshafts rotate at the same speed and the planet gears do not rotate.

If the vehicle is turning, the outer wheel will be forced to rotate faster than the inner wheel by having a greater distance to travel. The differential senses the torque difference between the sun gears. The planet gears rotate on their axes to allow the outer wheel to rotate faster than the inner one.

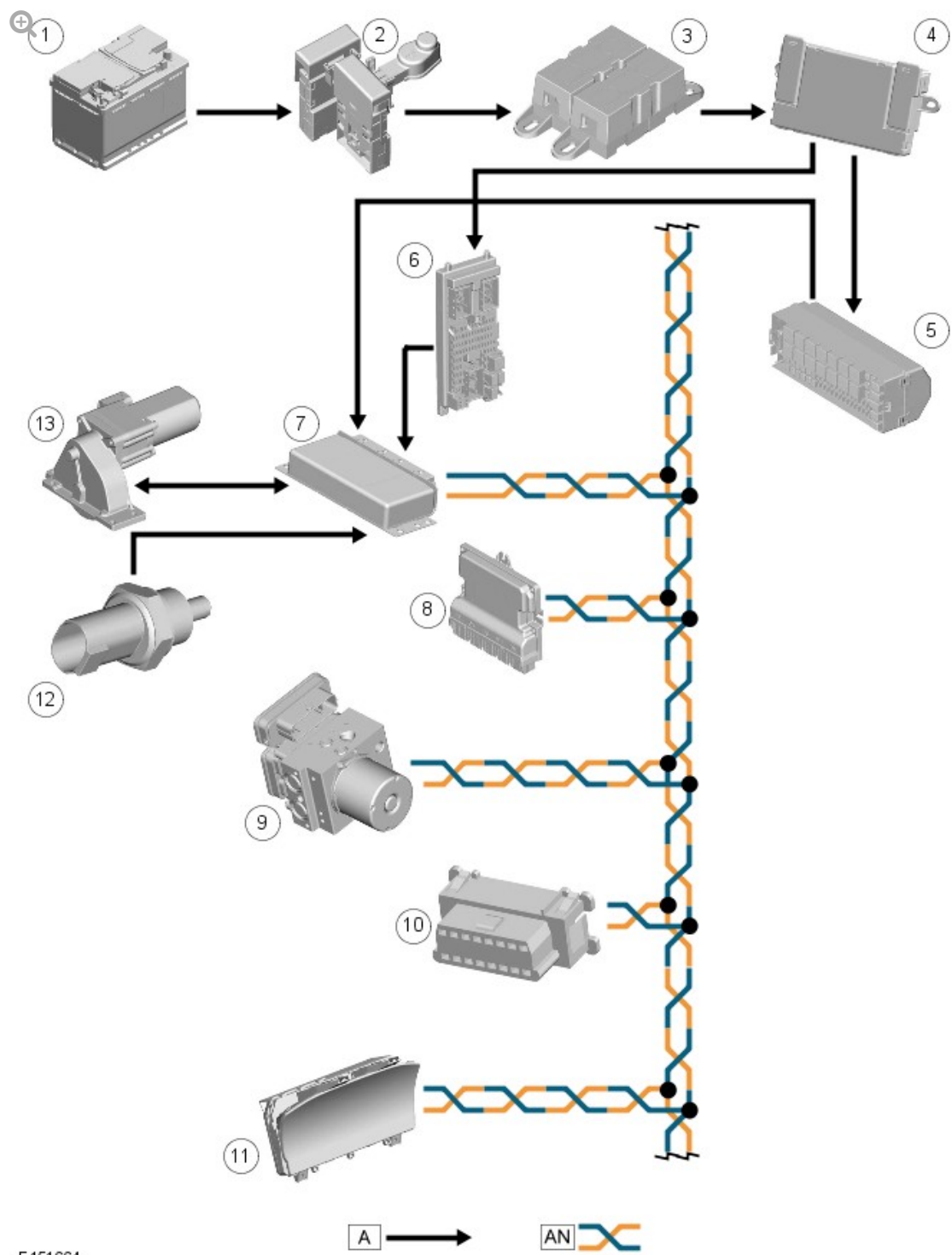
ELECTRONIC DIFFERENTIAL

The multi-plate clutch prevents excessive differential slip and therefore maximizes the traction performance of the vehicle. This is fundamentally different from 'braked' traction control systems, which can only counteract differential slip when it occurs.

A certain amount of differential slip is required to allow the vehicle to turn corners and to remain stable under control of the ABS (anti-lock brake system). The system is completely automatic and does not require any special driver input.

The multi-plate clutch actively controls the torque flow through the differential and optimizes the torque distribution in the driveline. The clutch biases the torque from the differential to the wheel with the higher grip and prevents the wheel with the lower grip from spinning.

CONTROL DIAGRAM



E151664

A = HARDWIRED; AM = HIGH SPEED CAN CHASSIS SYSTEMS.

ITEM	DESCRIPTION
1	Battery
2	Megafuse
3	Battery Junction Box 2 (BJB2)
4	Battery Junction Box (BJB)

5	Rear Junction Box (RJB)
6	Central Junction Box (CJB)
7	Rear Differential Control Module (RDCM)
8	Integrated Suspension Control Module (ISCM)
9	Anti-Lock Brake System (ABS) control module
10	Diagnostic connector
11	Instrument Cluster (IC)
12	Oil temperature sensor
13	Differential locking motor